

Foundations of Mycology & Anti-fungal Agents

John F. Fisher, MD + Daniel Anderson, PharmD, BCIDP • two-deck combined lecture companion

6 official objectives

2 ScholarRx bricks

Mycology morphology

Reproduction & spores

Antifungal MOA + toxicity

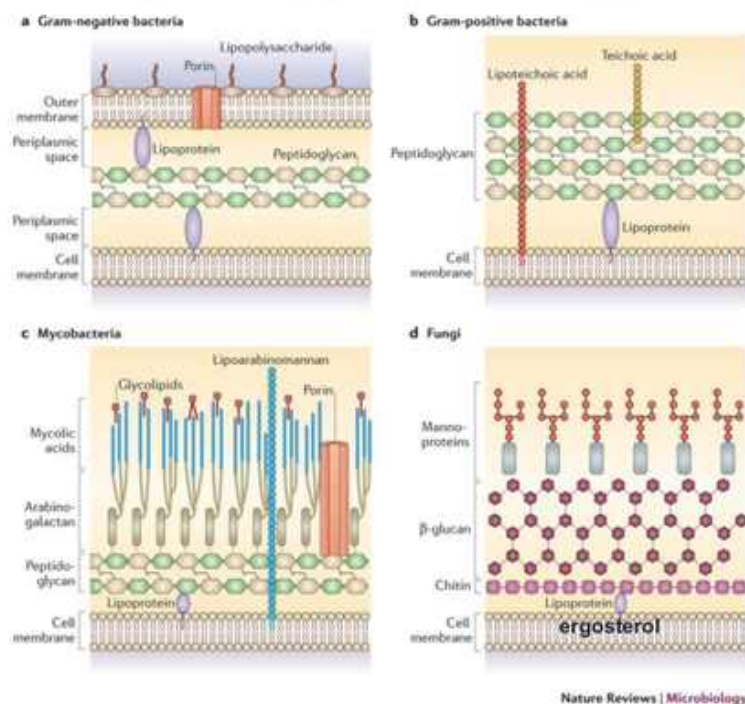
CURRENT COURSE STRUCTURE: this is one Core Lecture 7. The Fisher *Foundations in Mycology* deck and Anderson *Antifungal Agents* deck are co-primary halves of the lecture. The uploaded handouts are labeled “Lecture 10A” and “10B,” but those labels come from the handout source and do *not* control the current Week 8 numbering.

Official Week 8 Objectives

1. List five differences between fungi and bacteria.
2. Describe the various types of hyphae.
3. Distinguish between sexual and asexual reproduction in fungi.
4. Differentiate between sexual and asexual spores.
5. Classify important antifungals by mechanism of action.
6. Provide the most significant toxic side effects of various classes of antifungals.

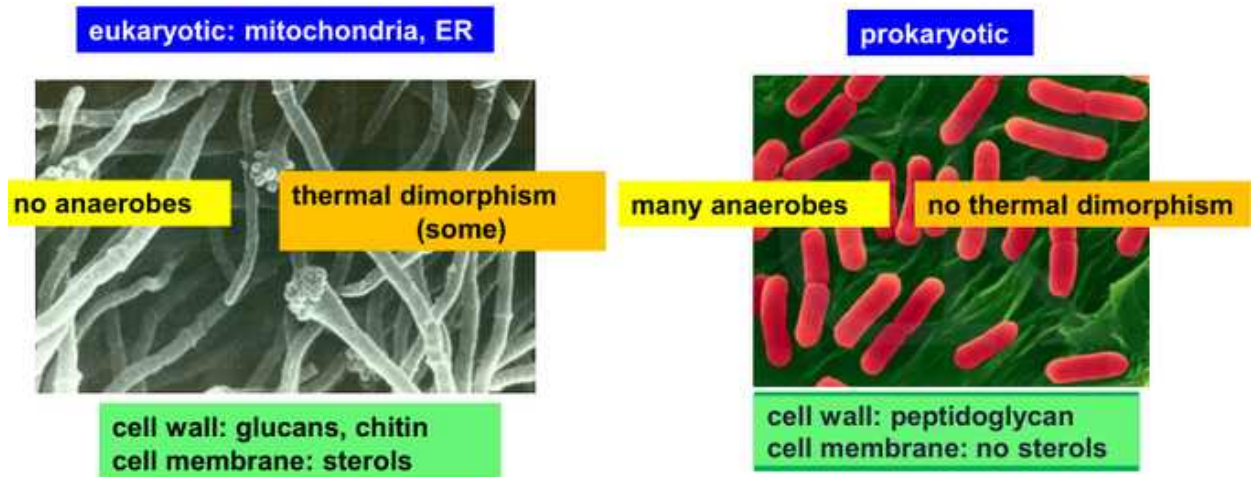
Part A - Fungal Foundations: Why Fungi Are Different

Microorganisms (comparisons)



Fisher mycology slide 3. The deck visually compares microbial envelopes and highlights ergosterol as a fungal membrane feature.

Fungi vs bacteria



Fisher mycology slide 4. The core fungi-vs-bacteria comparison: eukaryotic cellular machinery, glucan/chitin wall, sterol-containing membrane, limited anaerobiosis, and thermal dimorphism in some fungi versus bacterial peptidoglycan/prokaryotic biology.

Feature	Fungi - lecture framing	Bacteria - lecture framing
Cell type	Eukaryotic; mitochondria, ER, nucleus	Prokaryotic
Cell wall	Glucans + chitin	Peptidoglycan
Cell membrane	Sterols, especially ergosterol	No sterols in the lecture comparison
Anaerobiosis	Lecture emphasizes essentially no/rare anaerobes	Many anaerobes
Thermal dimorphism	Present in some fungi	Absent

THE FIVE-DIFFERENCE OBJECTIVE

Eukaryote • glucan/chitin wall • ergosterol membrane • little/no anaerobic growth in the lecture framework • some thermal dimorphism.

Handout enrichment: the 10A guide explicitly explains why these structural differences matter pharmacologically: fungi resemble human cells more than bacteria do, so the most useful selective targets are the **cell wall** and **ergosterol-containing membrane**.

Slides 5-10: Yeast, Mould, Dimorphism, and Hyphae

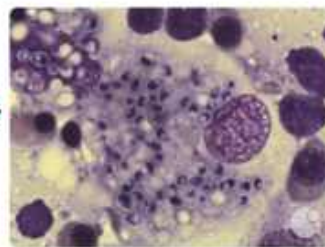
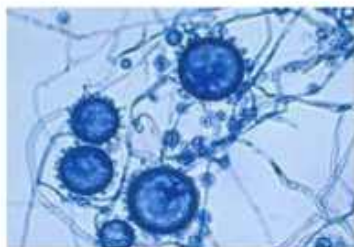
Fungi



yeast

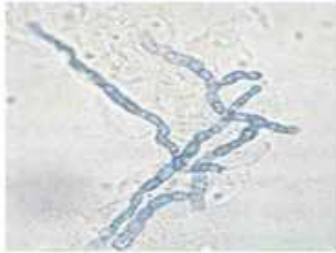


moulds



Fisher mycology slide 5. The lecture groups fungi into yeast, mould, and dimorphic forms.

Morphology



hyphae
(chain of cylindrical cells)



yeast
(single, budding cells)



hyphae & specialized structures

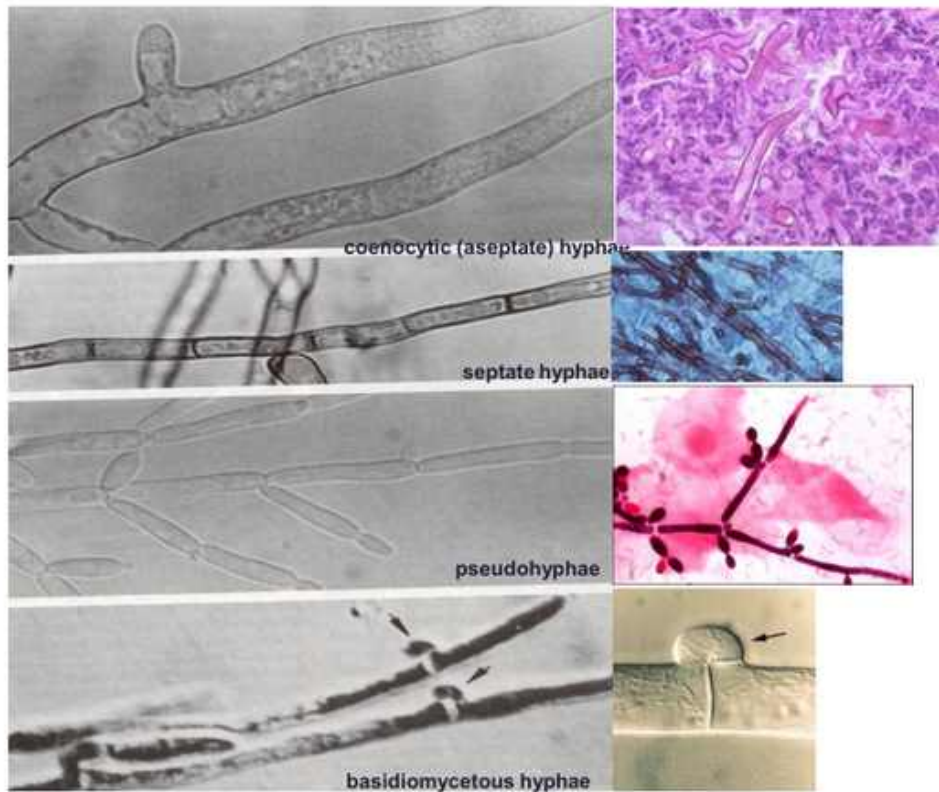


yeast & pseudohyphae

Fisher mycology slide 7. Morphologic vocabulary: hyphae are chains of cylindrical cells; yeasts are single budding cells; yeast can form pseudohyphae; moulds form hyphae and specialized structures.

Form	Recognition	Course connection
Yeast	Single budding cells	Candida and Cryptococcus are shown later as yeast examples
Mould	Filamentous hyphae; mycelium = mass of hyphae	Aspergillus / zygomycete morphology in classification slides
Dimorphic	Switches morphology with environment/temperature	Deck question expects yeast form at 37 C
Pseudohyphae	Elongated attached budding cells	Distinguished from true hyphae

Types of hyphae



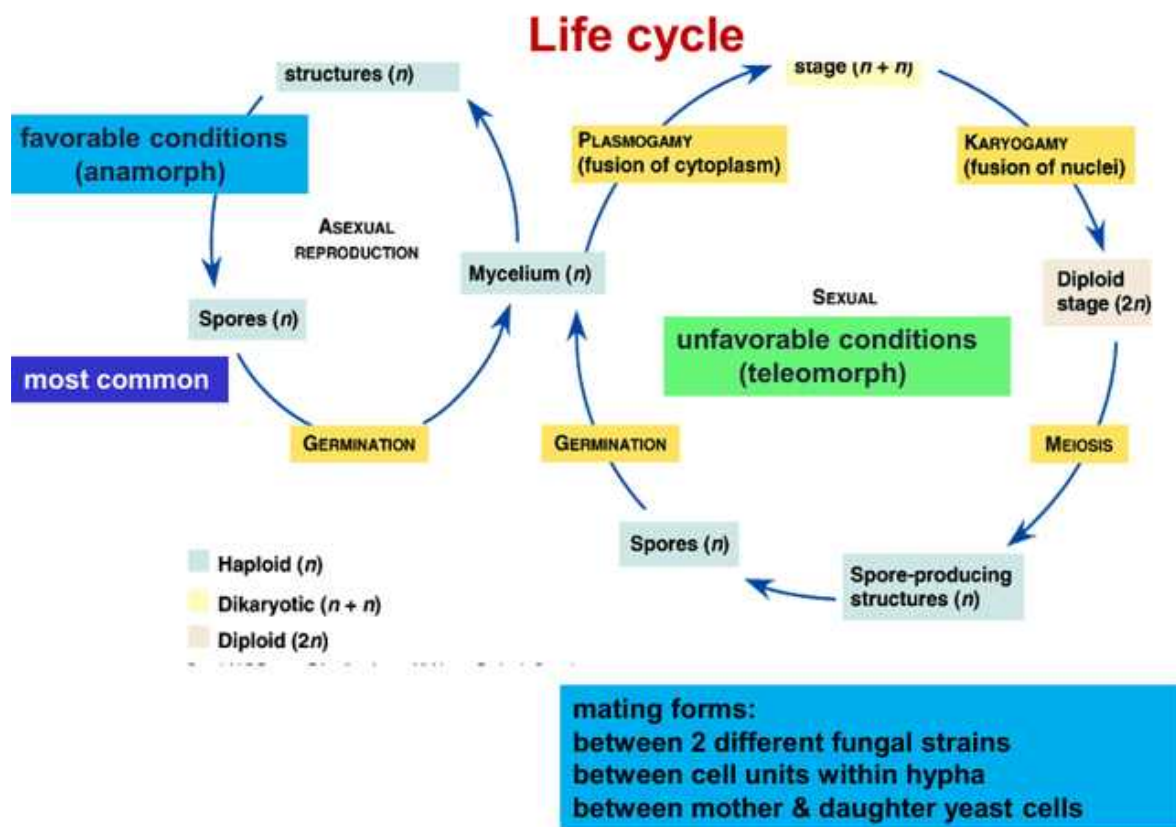
Fisher mycology slide 9. The deck directly contrasts coenocytic/aseptate, septate, pseudohyphal, and basidiomycetous hyphal forms.

TYPES OF HYPHAE

Coenocytic (aseptate) • septate • pseudohyphae • basidiomycetous hyphae.

Handout-only board bridge: 10A gives the mnemonic “**mould in the cold, yeast in the heat**” and connects coenocytic hyphae to *Mucor/Rhizopus* and septate hyphae to *Aspergillus*. Keep the mnemonic as enrichment; the deck itself tests the 37 C = yeast concept directly.

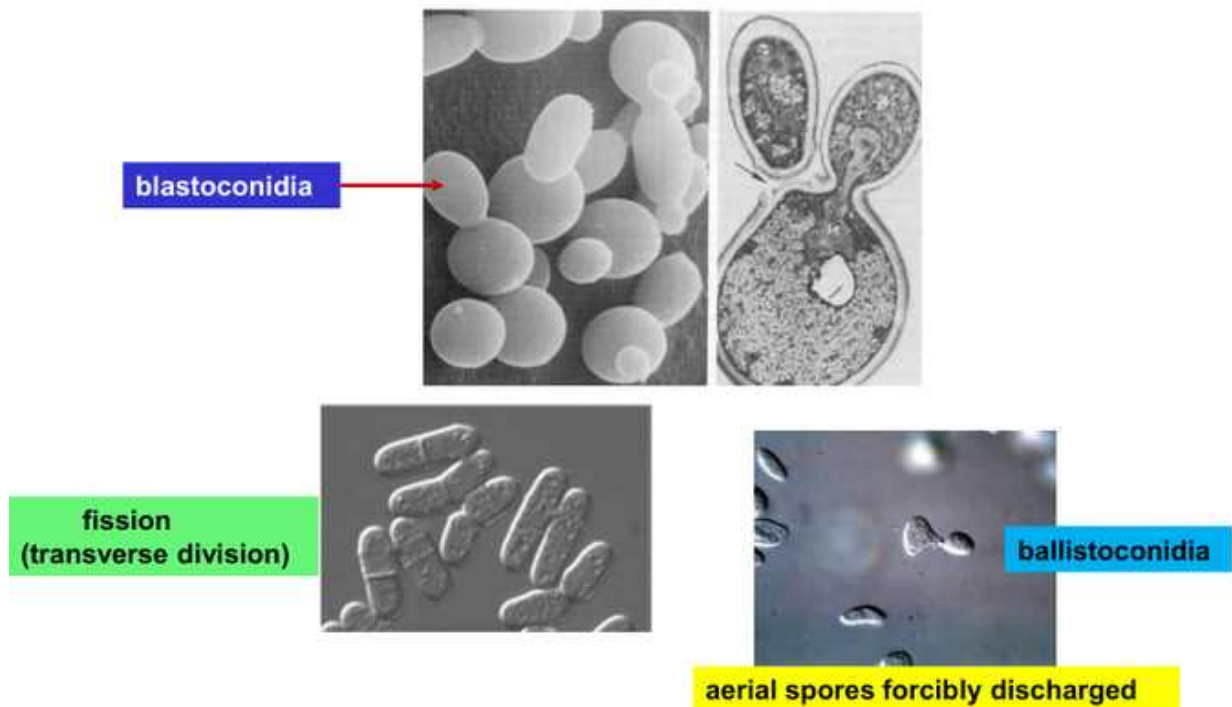
Slides 11-15: Sexual vs Asexual Reproduction



Fisher mycology slide 11. The life-cycle diagram separates favorable-condition asexual reproduction (anamorph) from unfavorable-condition sexual reproduction (teleomorph).

Asexual	Sexual
Favored under favorable environmental conditions	Lecture associates with unfavorable conditions
Mitosis	Plasmogamy → karyogamy → diploid stage → meiosis
Produces asexual spores / daughter cells	Produces sexual spores
Most common pathway in the deck life-cycle figure	Requires mating biology: compatible strains or self-compatible forms

Asexual reproduction--yeasts



Fisher mycology slide 15. Asexual yeast reproduction examples: blastoconidia, fission, and ballistoconidia.

SPORE DISTINCTION

Asexual spores arise through mitotic reproduction. Sexual spores follow nuclear fusion and meiosis.

10A enrichment: conidia are external asexual spores, sporangiospores are produced inside a sporangium, arthrospores arise from hyphal fragmentation, and blastospores/ blastoconidia arise by budding. These labels help decode the classification slides, but not every term is written out in the main deck.

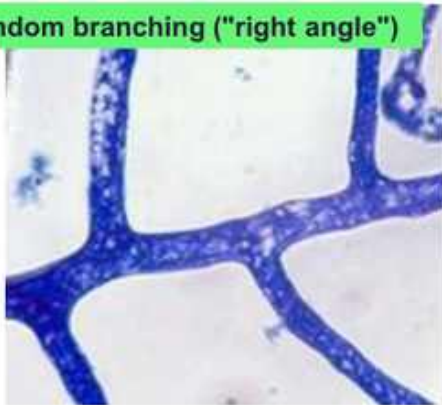
Slides 16-24: Course Classification & Morphologic Examples

Zygomycetes (asexual stage)

Mucor, Rhizopus, Rhizomucor
(most important medically)



random branching ("right angle")



sporangiospores (asexual)

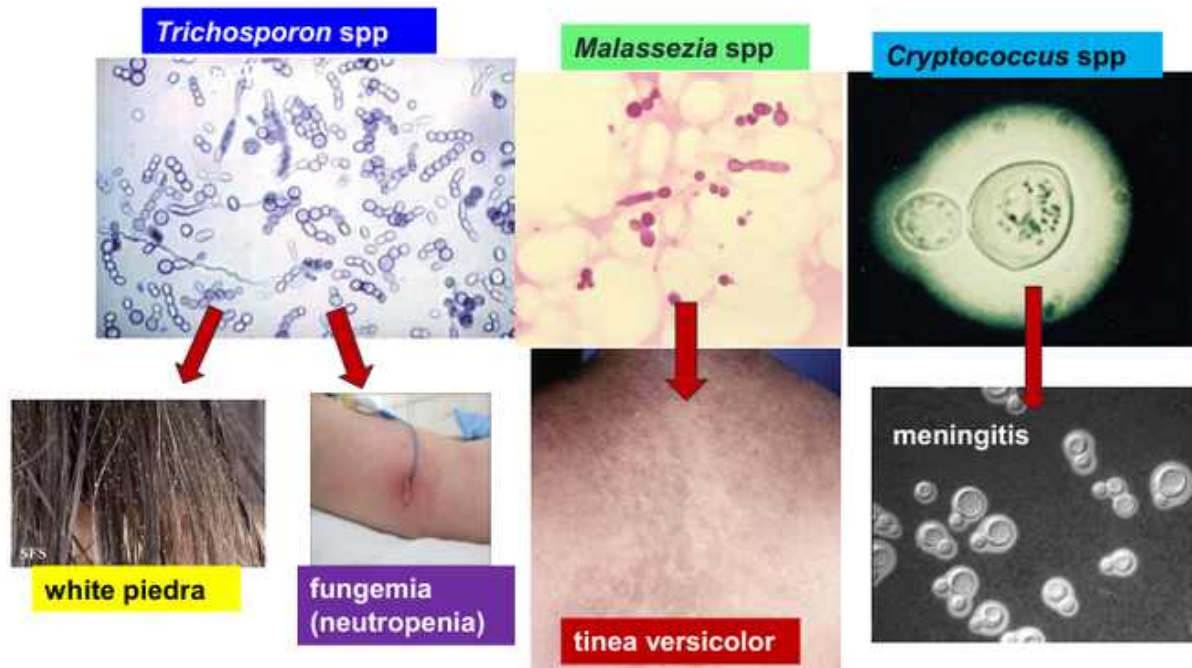
sporangium

sporangiophore



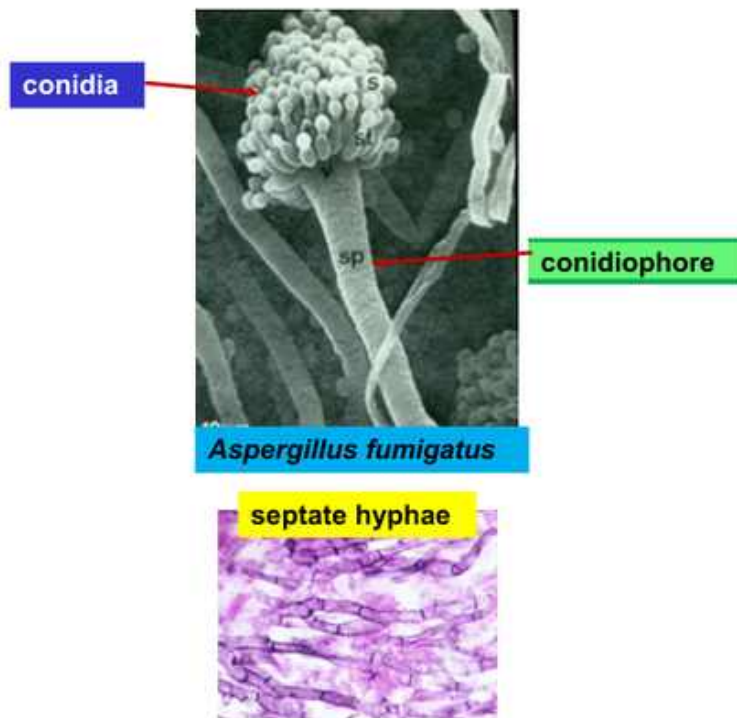
Fisher mycology slide 17. Course terminology: Zygomycetes include *Mucor*, *Rhizopus*, and *Rhizomucor*; the asexual stage shows sporangia/sporangiophores and random/right-angle branching.

Basidiomycetes (asexual stage)



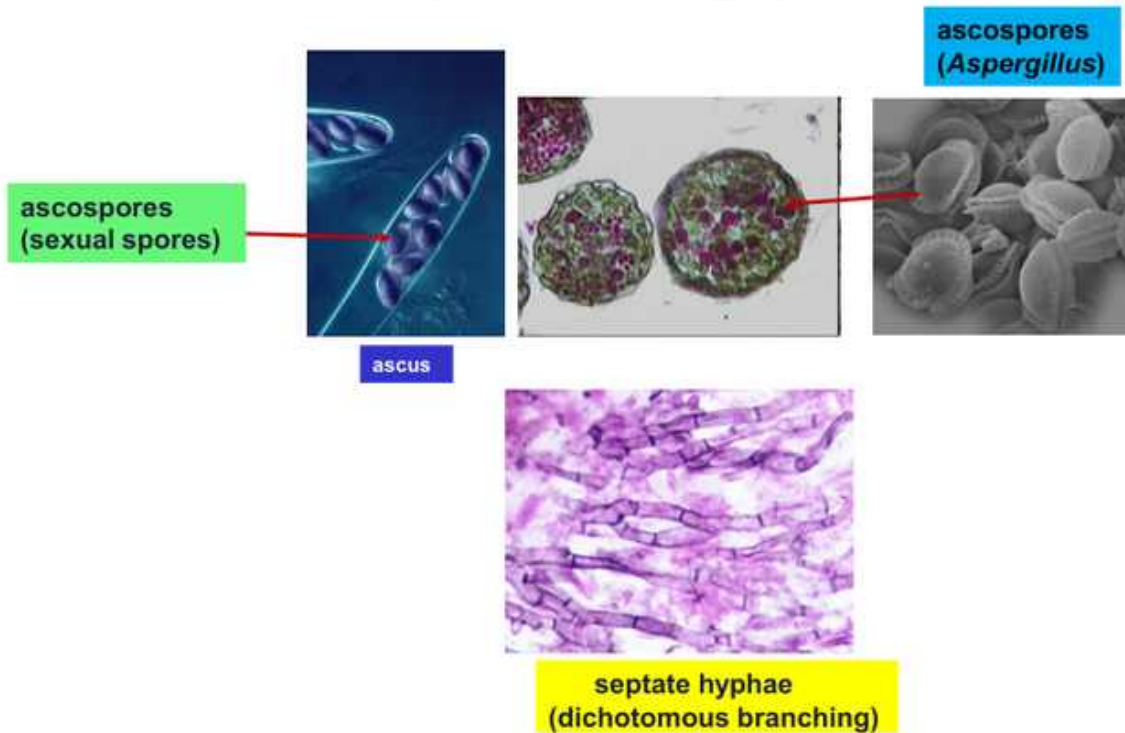
Fisher mycology slide 19. Basidiomycete examples in the deck include Cryptococcus, Malassezia, and Trichosporon with associated clinical labels.

Ascomycetes (asexual stage)



Fisher mycology slide 21. Ascomycete asexual stage: conidia on conidiophores; *Aspergillus fumigatus* with septate hyphae.

Ascomycetes (sexual stage)



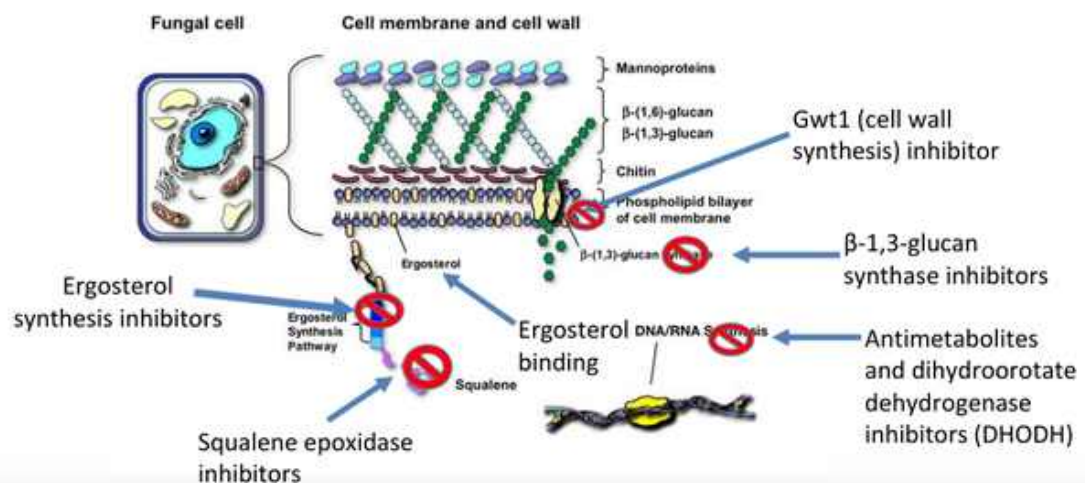
Fisher mycology slide 22. Ascomycete sexual stage: ascus and ascospores; deck also highlights septate hyphae with dichotomous branching.

Lecture group	Key morphology	Examples shown
Chytridiomycetes	Aquatic; swimming zoospores	Deck labels them non-pathogenic for humans
Zygomycetes	Coenocytic/aseptate; sporangiospores; zygospore sexual stage; random/right-angle branching	Mucor, Rhizopus, Rhizomucor
Basidiomycetes	Basidium → basidiospores; basidiomycetous hyphae	Cryptococcus, Malassezia, Trichosporon
Ascomycetes	Septate; conidia asexual; ascus/ascospores sexual; dichotomous branching shown	Aspergillus fumigatus

Use the course's terminology for this lecture. The uploaded handout also discusses “Deuteromycetes/Fungi Imperfecti” as a holding category. That category is *handout-only* and is not a dedicated classification slide in the current Fisher deck.

Part B - The Fungal Cell Becomes a Drug-Target Map

The Fungal Cell



Anderson antifungal slide 2. The antifungal deck turns the same fungal structures into pharmacologic targets: ergosterol synthesis, ergosterol binding, β -1,3-glucan synthesis, squalene epoxidase, nucleic-acid/pyrimidine pathways, and Gwt1.

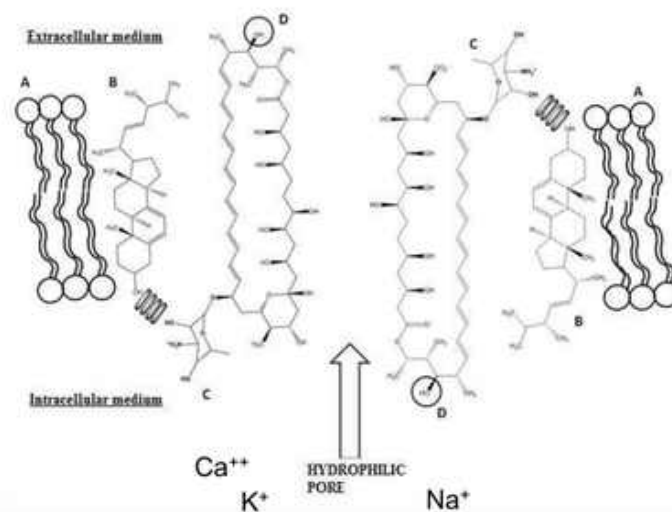
Fungal target	Drug/class	Mechanistic consequence
Ergosterol itself	Polyenes	Bind membrane sterol → pore formation / depolarization
14 α -demethylase	Azoles	Block ergosterol synthesis
Squalene epoxidase	Terbinafine	Early ergosterol-synthesis blockade
β -1,3-D-glucan synthase / Fks	Echinocandins , ibrexafungerp	Disrupt fungal cell-wall glucan synthesis
DNA/RNA metabolism	Flucytosine , olorofim	Disrupt nucleic-acid synthesis through different pathways
Gwt1	Fosmanogepix	Disrupt anchored proteins used for mucosal/epithelial adhesion

Ozzy checkpoint - click him when you can point to a fungal structure and name the antifungal class that targets it.

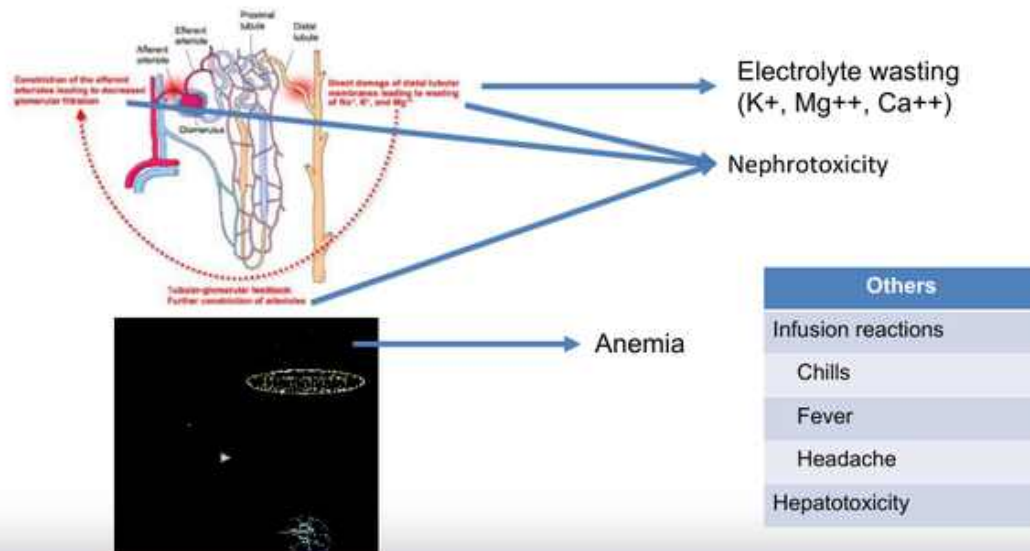
Deck wording mismatch worth noticing: antifungal slide 4 labels ergosterol/ cholesterol as “cell wall” sterols, while the deck’s fungal-cell target map and pore mechanism place the sterol target in the **cell membrane**. This companion follows the mechanistic diagrams and flags the inconsistent wording rather than silently blending it.

Polyenes: Amphotericin B & Nystatin

Mechanism of Action



Amphotericin Side Effects



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Anderson antifungal slide 6. Amphotericin toxicity slide: nephrotoxicity, electrolyte wasting, anemia, infusion reactions, and hepatotoxicity are highlighted.

AMPHOTERICIN B

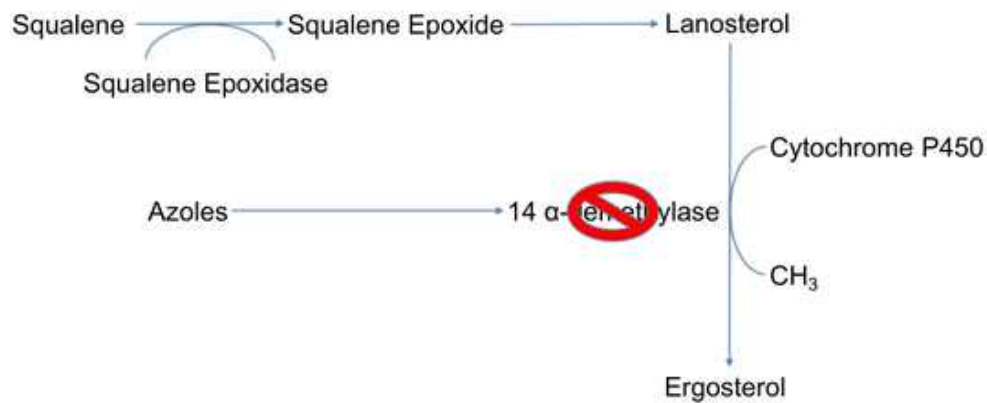
Bind ergosterol → membrane pores. Major exam toxicity = **kidney**, with K/Mg/Ca wasting highlighted by the deck.

- Additional deck toxicities: anemia; chills/fever/headache infusion reactions; hepatotoxicity.
- Deck spectrum: Candida, Aspergillus, Mucor, dimorphic fungi, Cryptococcus, other molds.
- Lipid formulations are shown as improving CNS penetration; the summary handout also notes reduced nephrotoxicity.

10B enrichment: nystatin has the same polyene mechanism but is used locally/topically rather than for systemic disease. The handout also uses the classic “Ampho-terrible” mnemonic; useful, but not faculty-slide wording.

Azoles: Block 14 α -Demethylase

Ergosterol Synthesis



Class-wide Side Effects

Hepatotoxicity

Cytochrome
P450
Inhibition

GI side
effects

QTc
prolongation*

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Anderson antifungal slide 11. Class-wide adverse effects: hepatotoxicity, CYP450 inhibition, GI effects, and QTc prolongation, with an asterisk because isavuconazole is the exception.

AZOLE RULE

14 α -demethylase inhibition → ↓ **ergosterol**. Class concerns = **liver + CYP interactions + GI + QTc**; isavuconazole **shortens** QTc.

Voriconazole

Spectrum of Activity	Pharmacological Considerations
Candida spp Aspergillus Scedosporium Fusarium Dematiaceous molds	Visual disturbances (color changes) Hallucinations and nightmares Photosensitivity Higher hepatotoxicity rates More potent CYP450 inhibitor Requires therapeutic drug monitoring

Anderson antifungal slide 13. Voriconazole: Aspergillus/Candida plus other molds; visual color changes, hallucinations/nightmares, photosensitivity, higher hepatotoxicity, strong CYP inhibition, and therapeutic drug monitoring.

Itraconazole

Spectrum of Activity	Pharmacological Considerations
Dimorphic fungi Aspergillosis Onychomycosis	Solution generally > capsule Capsule needs an acidic environment Solution should be taken on an empty stomach Requires therapeutic drug monitoring

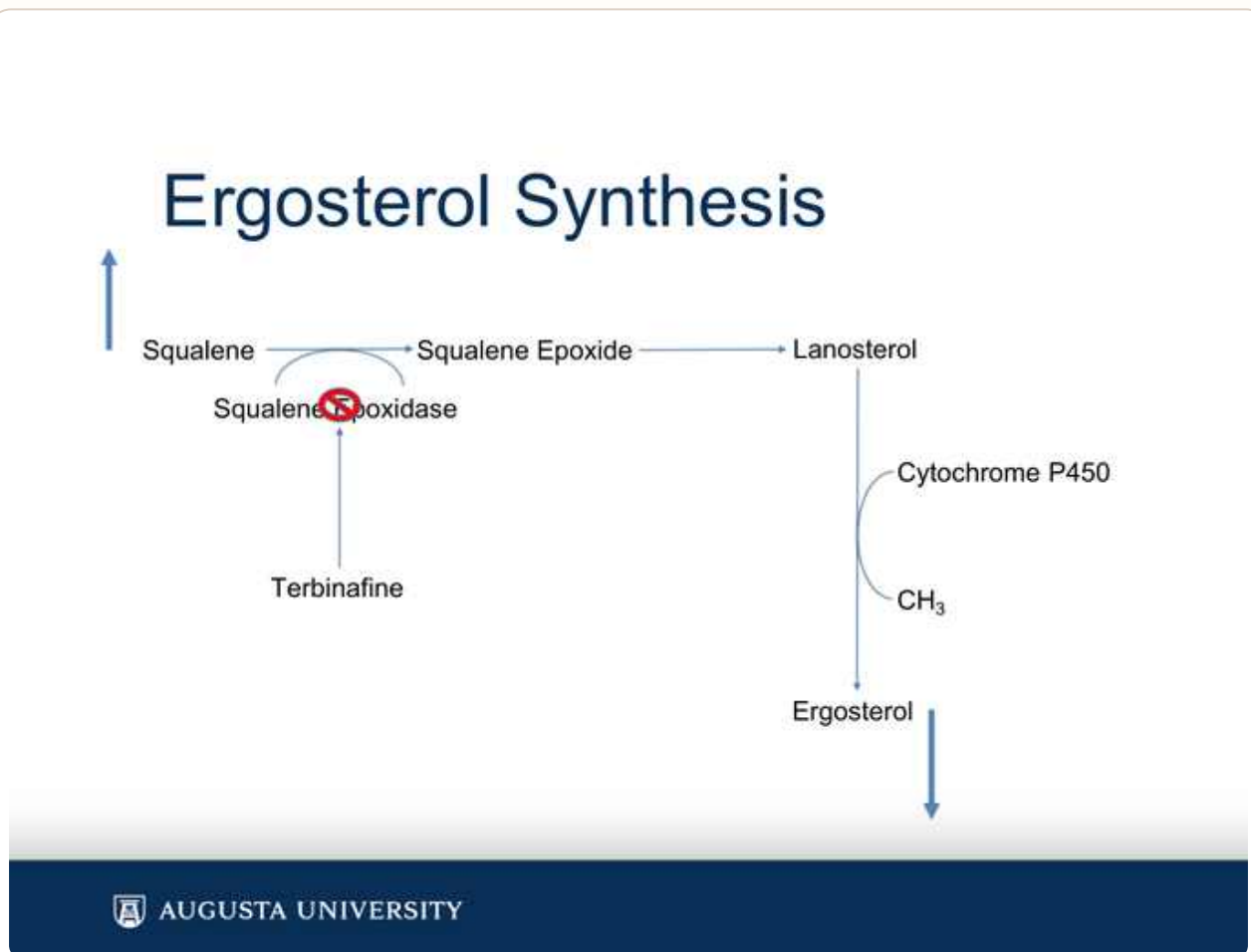
Isavuconazole

Spectrum of Activity	Pharmacological Considerations
Candida* Aspergillosis Mucormycosis	Shortens QTc interval LFT elevations reversible upon stopping therapy

Anderson antifungal slide 16. Isavuconazole: Aspergillus and mucormycosis emphasis; shortens QTc; reversible LFT elevations after stopping therapy.

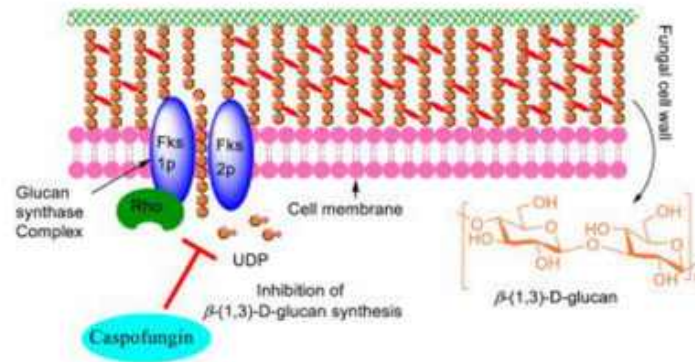
Azole	Deck activity emphasis	Deck practical / toxicity clue
Fluconazole	Candida, Cryptococcus, tinea, dimorphic fungi	Class-wide toxicities/interactions apply
Voriconazole	Candida, Aspergillus, Scedosporium, Fusarium, dematiaceous molds	Visual changes; hallucinations/nightmares; photosensitivity; TDM; potent CYP inhibitor
Posaconazole	Candida, Aspergillus, dimorphic fungi, Mucorales	Take oral form with food; prophylaxis; TDM; GI effects; electrolyte abnormalities
Itraconazole	Dimorphic fungi, aspergillosis, onychomycosis	Solution generally > capsule; capsule needs acid; solution empty stomach; TDM
Isavuconazole	Aspergillosis, mucormycosis; Candida marked with * on slide	Shortens QTc ; LFT elevations reversible after stopping

Other Mechanisms: Terbinafine, Echinocandins, Flucytosine, Newer Agents



Anderson antifungal slide 18. Terbinafine inhibits squalene epoxidase early in ergosterol synthesis; the deck primarily connects it to dermatologic/tinea indications.

Mechanism of Action



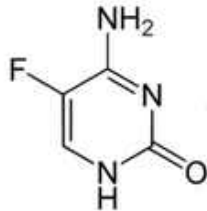
Anderson antifungal slide 20. Echinocandin mechanism slide: inhibition of β -1,3-D-glucan synthesis in the fungal cell wall.

ECHINOCANDINS

Micafungin • caspofungin • anidulafungin • rezafungin → inhibit Fks/ β -1,3-D-glucan synthase. Deck coverage is strongest for Candida; Aspergillus is marked with an asterisk; Mucor/Cryptococcus/Blastomycosis are poor/no coverage.

Flucytosine

Flucytosine (5FC) outside of cell

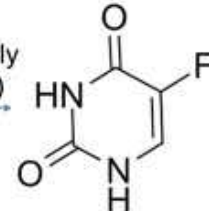


Permease (only
in fungal cells)

5FC inside cell

Deaminase (only
in fungal cells)

Fluorouracil (5FU)



Inhibition of protein synthesis

Inserted into RNA

Inhibition of DNA synthesis

dUMP and dTMP

5-fluorouridine
monophosphate

Anderson antifungal slide 23. Flucytosine is transported into fungal cells and converted to 5-FU, disrupting both RNA/protein synthesis and DNA synthesis.

Flucytosine

Spectrum of Activity	Adverse Drug Events	Pharmacological Considerations
• Candida • Cryptococcus	• Bone marrow suppression • Hepatitis	• Oral only • Used in combination • Dose based on ideal body weight

Anderson antifungal slide 24. Flucytosine summary: Candida/Cryptococcus; bone-marrow suppression and hepatitis; oral only; used in combination.

Agent	Target / MOA	High-yield lecture point
Terbinafine	Squalene epoxidase	Dermatologic / tinea emphasis
Echinocandins	Fks → β -1,3-D-glucan synthase	Cell-wall target; Candida emphasis
Ibrexafungerp	β -1,3-D-glucan synthesis at a different Fks binding site	Oral bioavailability, tissue penetration, less impacted by FKS mutation
Flucytosine	Fungal conversion to 5-FU → RNA + DNA synthesis inhibition	Bone marrow suppression; hepatitis; used in combination
Olorofim	DHODH → pyrimidine biosynthesis	Aspergillus/dimorphic fungi/some Fusarium; poor Candida/Mucorales
Fosmanogepix	Gwt1 inhibition	Blocks anchored adhesion proteins; broad experimental spectrum shown

Lower-priority / clinical-rotation material: the Anderson summary table explicitly says an asterisk marks information intended for clinical rotations. Ibrexafungerp, olorofim, fosmanogepix, and rezafungin appear in that newer-agent/informational layer. Know the mechanism map, but prioritize the classic classes and toxicities for the official objectives.

10B enrichment: flucytosine + amphotericin B for severe cryptococcal meningitis is added by the handout. The current deck itself says flucytosine is oral, used in combination, and active against Candida/Cryptococcus, without making that combination the central slide wording.

Official Objective 6: Toxicity Map

Class / drug	Most important toxicity / practical issue from the Anderson deck
Amphotericin B	Nephrotoxicity ; electrolyte wasting (K, Mg, Ca); anemia; infusion chills/fever/headache; hepatotoxicity
Azoles - class	Hepatotoxicity; CYP450 inhibition/interactions; GI effects; QTc prolongation
Voriconazole	Visual color changes; hallucinations/nightmares; photosensitivity; higher hepatotoxicity; TDM
Posaconazole	GI effects; low K/Mg/Ca; TDM
Isavuconazole	LFT elevation; shortens QTc instead of prolonging it
Echinocandins	Summary handout emphasizes hepatotoxicity; poor urinary/brain/eye penetration
Flucytosine	Bone marrow suppression ; hepatitis

IF YOU ONLY MEMORIZE THREE TOXICITIES

Amphotericin → kidney/electrolytes. **Azoles** → liver/CYP/QT. **Flucytosine** → bone marrow.

Antifungal Agents Handout

* indicates informational only and for use on clinical rotations

Class/Drug	Mechanism of Action	Spectrum of Activity	Notable Gaps in Coverage	Unique Adverse Drug Events	Monitoring/Miscellaneous
Amphotericin B	Creates a hydrophilic pore in the cell wall causing depolarization	Candida spp. Aspergillus Mucor Dimorphic fungi Cryptococcus	Candida auris Candida lusitanae Aspergillus terreus Fusarium Scedosporium	Nephrotoxicity Electrolyte abnormalities (K, Ca, Phos, Mg) Infusion related reactions Bone marrow suppression Hepatotoxicity	Monitor kidney function and electrolytes daily Pre- and post-hydration fluid Acetaminophen and diphenhydramine prior to administration Lipid formulations less nephrotoxic
Azoles					
Fluconazole	Inhibit ergosterol synthesis through inhibition of 14 α -demethylase	Candida spp Cryptococcus Tinea Dimorphic fungi	Candida krusei Molds	Hepatotoxicity GI side effects QTc prolongation (except isavuconazole) Visual disturbances (voriconazole) Hallucinations (voriconazole)	
Voriconazole		Aspergillus Candida spp Scedosporium Fusarium Dematiaceous molds	Mucor		Requires therapeutic drug monitoring More potent CYP inhibitor than the others
Itraconazole		Dimorphic fungi Aspergillus Onychomycosis	Mucor		Take capsule with food or acidic beverage Take solution on an empty stomach Requires therapeutic drug monitoring
Posaconazole		Candida spp Aspergillus Dimorphic fungi Mucorales	Fusarium Scedosporium		Take with food Requires therapeutic drug monitoring
Isavuconazole		Candida spp Aspergillus Mucormycosis			Shortens QTc interval Should not be used for bloodstream infections
Echinocandins Micafungin Caspofungin	Binds to Fks proteins on cell membrane to inhibit 1,3- β -D-Glucan synthesis	Candida Aspergillus*	Mucor Cryptococcus Blastomycosis	Hepatotoxicity	Poor urinary, brain, and eye penetration

Anderson antifungal slide 31. The deck's dense summary handout consolidates mechanisms, spectra, gaps, adverse effects, and monitoring. The asterisk note identifies selected clinical-rotation-only details.

Vignette / Practical Recognition

If the question gives...	Think...	Why
Eukaryote + glucan/chitin wall + ergosterol membrane	Fungus	Core structural comparison
Broad aseptate/coenocytic hyphae; right-angle/random branching in course language	Zygomycete / Mucor-Rhizopus group	Fisher slide 17
Septate dichotomous branching + conidia	Ascomycete / Aspergillus pattern	Fisher slides 21-22
Drug binds ergosterol and creates pores + renal toxicity	Amphotericin B	Polyene mechanism + toxicity
14 α -demethylase inhibition + CYP interactions	Azole	Ergosterol-synthesis blockade
Visual disturbances/hallucinations	Voriconazole	Specific slide toxicity
QT interval gets shorter	Isavuconazole	Exception to azole QTc pattern
β -1,3-D-glucan synthase/Fks inhibition	Echinocandin	Cell-wall synthesis target
Cryptococcus/Candida agent + marrow suppression	Flucytosine	5-FU pathway + hematologic toxicity
Tinea / dermatologic indication + squalene epoxidase	Terbinafine	Early ergosterol-synthesis blockade

ONE-LINE FINAL MAP

Fungal biology tells you the target: ergosterol membrane → polyenes/azoles/terbinafine; glucan wall → echinocandins; fungal nucleic-acid metabolism → flucytosine.

Source hierarchy: Fisher 29-slide mycology deck + Anderson 32-slide antifungal deck are co-primary. Uploaded 10A/10B handouts are secondary enrichment and do not control current lecture numbering or slide numbering.